

LEVEL 3 — HARD

Q1

RECURSIVE



Q2

FUNCTIONS AND POINTERS



Write a function `int removeElement(int *arr, int *n, int value)` that removes all occurrences of value from the array `arr` using pointers and returns the new length of the array.

Sample Input:

Enter the size of the array: 6

Enter the elements of the array: 1 2 3 4 3 6

Enter the value to remove: 3

Sample Output:

Array after removing 3: 1 2 4

New length of the array is 4

2 / 2 saved

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int removeElement(int*a,int *n,int v){
4     int j=0;
5     for(int i=0;i<*n;i++)
6         if(a[i]!=v)
7             a[j++]=a[i];
8     *n=j;
9     return j;
10 }
11 int main(){
12     int n,v,a[100];
13     cin>>n;
14     for(int i=0;i<n;i++)
15         cin>>a[i];
16     cin>>v;
17     n=removeElement(a,&n,v);
18     for(int i=0;i<n;i++)
19         cout<<a[i]<<" ";
20     cout<<"\n New length:"<<n;
21     return 0;
22 }
```

Output

```
1 2 4 6
New length:4
```

Result Available
Your score: 96 / 100

QUESTIONS

LEVEL 3 — HARD

Q1

recursive

Q2

functionsandpointers

2 / 2 saved

Level 3 — Hard palindrome (H) — recursive

Write a recursive function to check if a given string is a palindrome.

Sample Input:

Enter a string: radar

Sample Output:

"radar" is a palindrome.

Test Case 1: Enter a string: Hello

Test Case 2: Enter a string: Python

Test Case 3: Enter a string: Programming

Your Code

```
1 #include<iostream>
2 using namespace std;
3 bool palindrome(string s,int l,int r){
4     if(l>=r) return true;
5     if (s[l]!=s[r]) return false;
6     return palindrome(s,l+1,r-1);
7 }
8 int main(){
9     string s;
10    cout<<"enter a string:";
11    cin>>s;
12    cout<<(palindrome(s,0,s.length()-1)?"palindrome":"not
13    return 0;
14 )
15
```

Output

enter a string: not palindrome

Submitted